

Grounding the Loop on Both Sides: Interaction as a Third Test-Time Compute Axis, and Why Its Gains Are Invisible Without Grounded Evaluation

Bojie Li
Pine AI
boj@19pine.ai

Abstract

Test-time compute is dominated by two scaling axes that both operate inside a model’s own token space: *reasoning* (longer chains of thought) and *sampling* (best-of- N). Both are bounded, by the data-processing inequality (DPI), by the information already in the weights and prompt. A third axis—*interaction* with an environment that returns grounded feedback—can exceed that bound by importing information from outside the model. We argue that a single variable, **grounding**, governs this axis, and that it must hold on *both* sides of the loop: grounded *feedback* drives improvement past the reasoning/sampling ceiling, and grounded *evaluation* is required even to *measure* the improvement. We formalize the first claim as a data-processing-inequality argument over a Type 0–3 feedback taxonomy. For the second, we show that on rendered visual artifacts the default vision–language-model (VLM) judge is structurally blind to the geometric defects the loop fixes—it rates 14 of 15 broken academic figures “perfect”—and that a VLM *reviewer* even makes layouts worse. Replacing it with a deterministic DOM-geometry instrument (overlap, clipping, overflow, and box-group misalignment, measured exactly from the rendered layout) reveals large, significant defect reductions under grounded feedback across **four** visual modalities measured identically: academic figures –74%, dense slides –73%, responsive web pages –47%, and SVG/CSS animations –40% (all paired sign-test $p < 0.002$; all bootstrap 95% CIs exclude zero). On execution-grounded code the harness lifts a frontier model from 66.7% to a strict 100% pass-rate, and an in-modality control isolates the *execution* of tests—not the extra reviewing pass—as the cause; this replicates across three model families and a held-out suite, and we reproduce the grounded-geometry result on a second proposer (Gemini 3.1 Pro) under the same model-free instrument. At a fixed 20K-token budget, reasoning-only and best-of- N saturate (73.3%/86.7%) while every interaction strategy reaches $\geq 97.8\%$, and the proposer–reviewer harness does so at the lowest token cost with zero seed variance. Finally, the harness’s behavior partially distills into an 8B student. Interaction scaling is real, distinct, and predictable—but only visible when both the feedback and the metric are grounded.

1 Introduction

Scaling test-time compute has become the dominant lever for improving large models on artifact-producing tasks—code, web pages, slides, figures, animations, video edits, research reports. The literature has formalized two axes, and both operate *inside the model’s own token space*. *Reasoning*

scaling [4, 17, 22, 23] spends more tokens thinking before committing to an answer. *Sampling scaling* [1, 21] draws more independent attempts and selects one. In both, every additional token is a function of the same frozen weights and the same fixed prompt: more compute reorganizes information already in the system but imports none.

A third axis breaks this closure: *interaction* with an external environment that returns **grounded** feedback—a test result, a rendered layout, a search verdict. Such feedback is not a function of the weights; it is an observation of the artifact’s real behavior, and it can carry information the model never had. This axis is well established empirically [6, 15, 16, 24], but accounts of *why* and *when* it should beat reasoning- or sampling-only scaling remain informal, and—we argue—half-stated. Practitioner reports at industrial scale sharpen the stakes: in a controlled 900-run deployment study at ByteDance, frontier coding models reach >80% functional *correctness* on a real product feature yet fail the delivery-quality dimensions (reliability, maintainability, performance) until a feedback “harness” is added—and it is the harness, not a larger model or more samples, that carries the gain [7]. That report also had to invent a multi-axis quality metric before the gap was even visible: a field observation of exactly the two-sided grounding problem we formalize below.

Grounding is the variable, on both sides of the loop. Our thesis is that one variable governs interaction scaling: **grounding**, the property that a signal is computed from the artifact’s actual behavior rather than from a model’s opinion of it. And grounding must hold on *both* sides of the proposer–reviewer loop (Figure 1):

1. **Grounded feedback** (the reviewer signal that drives revision): run the code and read `pytest`; render the HTML and measure the real bounding boxes; issue the query and read the result. This is the side prior work emphasizes, and a data-processing-inequality argument (Section 3) shows it is exactly what lets interaction exceed the ceiling that bounds reasoning and sampling.
2. **Grounded evaluation** (the metric that *measures* the gain): the same observation, used as the score. This side is almost always overlooked, and getting it wrong silently hides the effect.

The second point is not a detail. The default evaluator for visual artifacts is a vision–language model (VLM) reading a screenshot, and it is *structurally* blind to the defects that matter: content overflowing the canvas is cropped off-frame before the image reaches the judge, and small overlaps fall below its acuity. On dense academic figures this VLM judge rates 14 of 15 single-shot renders “perfect” (an initial 15-figure probe) when a deterministic check finds only 3 are actually clean (Section 6). Worse, when the same VLM is used as the *reviewer*, its edits make slide geometry *worse*. Ungrounded on either side, the third axis either does not fire or cannot be seen.

Contributions.

1. **A grounding framework (Section 3).** A Type 0–3 feedback taxonomy and a data-processing-inequality argument that motivates why ungrounded review and reasoning (and best-of- N selection) saturate while grounded feedback need not—plus the symmetric observation that ungrounded *evaluation* cannot measure the gain. Beyond the documented *biases* of model-as-judge evaluation [25], we isolate a sharper failure: for artifacts whose quality is a measurable physical property, a screenshot judge is *structurally blind* (not merely noisy), which silently hides a real interaction-scaling effect.
2. **Interaction is a distinct, unbounded axis (Section 5).** At a matched 20K-token budget, reasoning and best-of- N saturate (73.3%/86.7%) while every interaction strategy reaches $\geq 97.8\%$; an in-modality control isolates test *execution* as the active ingredient, and the harness is the most token-efficient, zero-variance strategy.

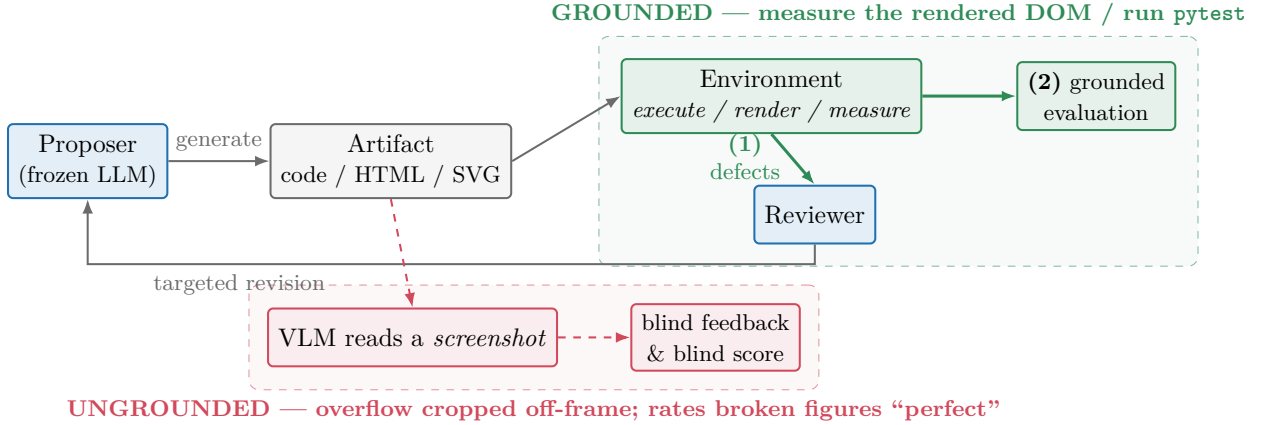


Figure 1: **Grounding must hold on both sides of the interaction loop.** A proposer emits an artifact; a grounded environment *executes/renders/measures* it, and that same observation does double duty: (1) as grounded *feedback*, it becomes a defect list the reviewer turns into a targeted revision—the channel that lets interaction import information from outside the weights and exceed the reasoning/sampling ceiling (Section 3); (2) as grounded *evaluation*, it is the score that makes the gain measurable. Substituting the default VLM-on-a-screenshot judge (red lane) breaks *both*: it is structurally blind to off-canvas overflow and fine overlap, so it neither scores the defect nor (as a reviewer) fixes it.

- Grounded evaluation, and a deterministic instrument (Section 6).** We show the VLM judge is structurally blind, build a deterministic DOM-geometry+alignment instrument, and report large, significant, CI-backed defect reductions under grounded feedback across four visual modalities measured identically—gains the standard metric cannot even see—plus a controlled ablation in which VLM feedback makes layout *worse*.
- Robustness and internalization (Sections 7 and 8).** The effects replicate across model families (including grounded geometry on Gemini 3.1 Pro), to held-out tasks, and under a propose-heavy budget allocation; and the harness’s behavior partially distills into an 8B student at $\sim 10\times$ lower deployment cost.

Availability. All code, task suites, prompts, the deterministic instruments, and an interactive results browser with per-task single-shot-vs-reviewed renders are available at <https://github.com/bojieli/interaction-scaling-paper>.

2 Related Work

Test-time compute scaling. Two axes are established. *Reasoning* scaling—chain-of-thought, tree search, and extended thinking [4, 17, 18, 22, 23]—spends more tokens before committing. *Sampling* scaling—self-consistency and best-of- N [1, 21]—draws multiple attempts and selects. Both transform the same frozen weights and prompt; we make precise (Section 3) the sense in which they are bounded, and position *interaction* as a third axis that imports external information.

Self-correction and critique. Iterative refinement with model-generated feedback—Reflexion [16], ReAct [24], Self-Refine [11], CRITIC [6]—is widely used, but its reliability is contested: Huang et al. [8] show models often cannot self-correct reasoning without an external signal, and Kamoi

et al. [9] chart when correction does and does not help. Our taxonomy makes the distinction explicit: *ungrounded* critique (Type 1) is bounded, whereas refinement against an executed/measured result (Type 3) is not—and our in-modality control (Section 5) shows it is the external execution, not the extra critique pass, that carries the gain.

Grounded and execution feedback. Tool use and execution feedback—Toolformer [14], Gorilla [12], Voyager [20], RLEF [5], and, for website generation specifically, WebGen-Agent’s multi-level visual/functional feedback [10]—and the thinking-vs-doing analysis of Shen et al. [15] establish that acting on an environment helps. We contribute the information-theoretic account of *why*, a feedback-actionability taxonomy that predicts *when*, and—crucially—the requirement that the *evaluation* be grounded too. We also compare harness architectures (single-agent loops [19], sample-and-select [13], proposer-reviewer) under a matched budget.

Model-as-judge reliability. LLM- and VLM-as-judge evaluation is now standard [25], and its failure modes—position, verbosity, and self-preference biases, and broader reward-model fragility [2]—are documented. Our finding is sharper and, to our knowledge, not previously isolated: for artifacts whose quality is a measurable *physical* property (the geometry of a rendered layout), a screenshot judge is not merely biased but *structurally blind* to defects cropped off-frame or below its acuity, so it both fails to score the defect and, used as a reviewer, fails to fix it. This is why a deterministic instrument is necessary, not merely preferable, to detect interaction scaling on visual modalities.

Practitioner evidence at scale. Industrial deployment independently corroborates the gap our framework targets. Hong [7] reports a controlled study at ByteDance—3 frontier coding models $\times$ 3 agent frameworks $\times$ 100 runs (900 total) on a single real product feature under a fixed prompt—in which functional *correctness* exceeded 80% for every model-framework pair, yet “deliverability” axes (UI usability, reliability, maintainability, performance, compatibility) collapsed to a failing 40–60% with high run-to-run randomness; only after adding harness “infrastructure”—context engineering, architectural constraints, and persisted team memory—did deliverability rise to $\sim$ 80% while correctness barely moved (80% $\rightarrow$ 90%). The same report notes that despite $>90\%$ of one team’s code being model-written, throughput rose only $1.6\times$, far short of the raw generation speedup—the macroscopic shadow of quality that single-shot generation does not deliver. This is the signature we formalize and measure: model and sampling scaling saturate the easy-to-see axis, the interaction loop carries the rest, and—tellingly—the report had to define a multi-axis quality metric before the gap was visible at all, an industrial analogue of our grounded-evaluation requirement (and its grounded fixes, e.g. browser-driven validation before commit, are Type-3 feedback in our taxonomy, not ungrounded self-review).

Information theory. The data-processing inequality [3] is the lens for our bound: any post-processing of the model’s outputs (reasoning, ungrounded review) cannot increase information about the target, whereas an environment observation introduces a new channel.

3 The Grounded-Feedback Framework

Our contribution is not a new self-correction method but an account of *why* interaction is a separate axis, *when* it helps (a feedback-actionability taxonomy), and a previously-missing requirement—that the *evaluation* be grounded too, without which the effect is unmeasurable.

3.1 A feedback taxonomy by grounding

We classify the reviewer’s feedback channel by how much external, task-conditional information it injects:

- **Type 0 – none.** Single-shot; no feedback.
- **Type 1 – ungrounded critique.** An LLM/VLM critiques the artifact from its weights (“looks wrong”). No execution, no measurement.
- **Type 2 – static analysis.** Linters/type-checkers: signal about the artifact’s *form*, not its behavior.
- **Type 3 – grounded.** The artifact is exercised and the result observed: **3a** execution (run tests, read tracebacks), **3b** rendered geometry (measure the DOM), **3c** temporal (sample animation frames), **3d** factual (search and verify claims).

3.2 Why grounding breaks the ceiling: a DPI bound

Let A^* be the (task-determined) correct artifact and A a candidate. The proposer is a channel from its inputs Θ (weights) and prompt X to A . A *reviewer* that only reads A and reasons forms the Markov chain $A^* \rightarrow (\Theta, X) \rightarrow A \rightarrow R$, where R is its critique. By the data-processing inequality,

$$I(A^*; R) \leq I(A^*; (\Theta, X)),$$

so *no* amount of reasoning or ungrounded review (Type 0–2) can recover more information about A^* than is already in (Θ, X) . This bounds reasoning-only scaling. Sampling is bounded differently but no less tightly: for any finite N , best-of- N can only *select* among the N candidates the proposer actually draws, so even an oracle (grounded) verifier cannot return a correct artifact the proposer is too unlikely to sample within the budget—its practical ceiling is the high-probability support of the proposer’s own output distribution, not a new information channel. Both ceilings appear empirically (Figure 2): on hard code, reasoning saturates at 73.3% and best-of- N (with an oracle verifier) at 86.7%. We stress that the inequality bounds the *information channel*, not achievable quality directly; we use it to *motivate* the saturation we then measure, and treat that measured saturation as the evidence.

Grounded feedback (Type 3) escapes both ceilings by feeding an environment observation $E = g(A, \text{world})$ —a test outcome, a measured bounding box, a search verdict—back into *generation*. Conditioning the next proposal on E lets the proposer synthesize a candidate it would not otherwise have sampled: formally $I(A^*; (R, E))$ can exceed $I(A^*; (\Theta, X))$ because E carries information obtained by *running* the artifact against reality. This is the crucial difference from selection (best-of- N cannot create a missing candidate) and from reasoning (which adds no information), and it is why interaction is a distinct axis rather than a constant-factor speedup.

The symmetric claim: ungrounded evaluation cannot measure the gain. The same inequality applies to the *scorer*. A metric M that is itself a function of (Θ, X) —e.g. a VLM judging a screenshot, whose view of A is a lossy, cropped image—satisfies $I(A^*; M) \leq I(A^*; (\Theta, X))$ and cannot certify an improvement that lives in the part of A it cannot observe. Grounding the evaluation (measure the artifact directly) is therefore not optional polish: it is a precondition for *detecting* interaction scaling on any modality whose quality is not fully visible to the default judge. Section 6 shows this is exactly the situation for rendered visual artifacts.

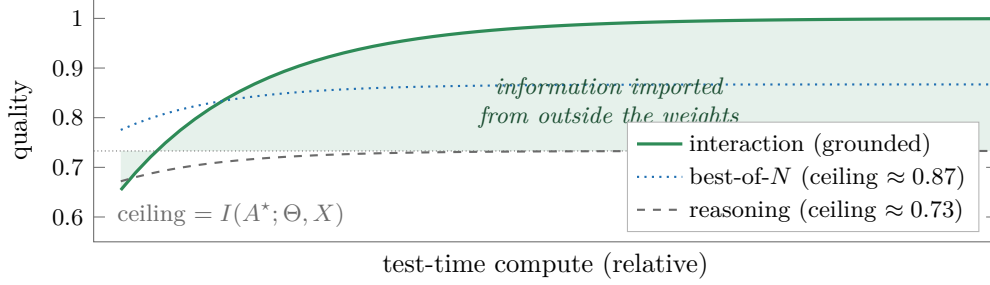


Figure 2: **The DPI ceiling.** Reasoning and best-of- N are functions of the frozen weights/prompt (Θ, X) and saturate at the information already present (measured 73.3% and 86.7% on hard code at a 20K-token budget; Section 5). Grounded interaction imports external information each cycle and exceeds the ceiling, reaching strict 100%. Schematic; saturation values are the measured ceilings.

3.3 Predictions

The framework makes three testable predictions, each tested below: **(P1)** reasoning- and sampling-only scaling saturate strictly below the interaction ceiling (Section 5); **(P2)** within interaction, more actionable grounded channels lift more, and the *execution* of the artifact—not the extra review pass—is the active ingredient (Section 5); **(P3)** on modalities whose quality is invisible to the default judge, the gain is only detectable with a grounded metric, and an ungrounded reviewer can even regress quality (Section 6).

4 Method: A Budget-Aware Proposer–Reviewer Harness

Architecture. The harness is pure scaffolding around a *frozen* frontier model (Claude Sonnet 4, `claude-sonnet-4-20250514`, temperature 0 unless noted): no fine-tuning, no retrieval, no learned controller. A **proposer** emits an artifact; an **environment** executes/renderers/measures it to produce a grounded signal; a **reviewer**—the same model in a diagnostic role—converts that signal into a typed defect list; the proposer revises. The loop repeats up to an iteration cap and *keeps the best-scoring iteration* (so the reviewed condition can never score below single-shot under the same metric). Single-shot is one proposer call.

The three grounded instruments. Every modality is paired with a deterministic or near-deterministic instrument that both *feeds back* and *scores*:

- **Execution** (pytest): exact pass/fail plus the failing assertion and traceback.
- **DOM geometry**: the artifact is rendered headless and every element’s true bounding box is read from the layout engine; we compute, exactly, text-on-text overlap (≥ 6 px), out-of-bounds/clipping, container overflow, document overflow (> 16 px), and—new here—*box-group misalignment* (rows/columns of card or pillar boxes flagged for unequal size, misaligned far edges, or uneven gutters). For scrollable web we score horizontal overflow at desktop (1920) and mobile (375) widths; for animations we probe the DOM at each sampled frame.
- **Native video** (Gemini 3.1 Pro): the rendered clip is judged whole against per-requirement binary checks—a near-grounded substitute where pixels, not a still, are observed.

Where no deterministic instrument exists (flowing web content, factual research), we fall back to a binary per-requirement rubric, flagged as such.

Table 1: Modality instantiations. Each modality pairs a grounded feedback channel with the instrument that both drives revision and scores the result. “†” marks the four modalities scored by the deterministic DOM-geometry instrument and analyzed jointly in Section 6.

Modality	Type	Grounded instrument	Hard suite (N)
Code	3a	<code>pytest</code> pass/fail + traceback	15 (+11 deep-spec)
Academic figures†	3b	DOM geometry + alignment	20
Slides†	3b	DOM geometry + alignment	12 real-paper
Web pages†	3b	DOM geometry, multi-width	20
Animations†	3c	DOM geometry, per-frame	20 SVG/CSS
Video editing	3a+3c	script exec + Gemini-native rubric	15
Deep research	3d	search-grounded factual rubric	15

Budget. A run has a token budget B split across proposer and reviewer calls; Section 7 sweeps the split. Table 1 lists the modality instantiations.

All artifacts are produced under a common *design-principle* generation prompt (proximity, alignment, repetition, contrast, deliberate color) so that single-shot quality is already strong; this makes the headroom we measure a property of genuine task difficulty rather than a weak prompt.

Evaluation protocol. Unless noted, the proposer runs at temperature 0 and each modality is evaluated over three independent on-policy seeds; the reviewed condition keeps the best-scoring iteration (cap ≤ 3 for geometry, ≤ 5 for code), so it can never score below single-shot under the same metric. The unit of comparison is the (task, seed) pair. For each result we report a two-sided paired sign test over decisive pairs and, for the geometric and code effects, a paired-bootstrap 95% confidence interval (10k resamples) on the effect size.

5 Interaction Is a Distinct, Unbounded Axis

Reasoning and sampling saturate; interaction does not (P1). On 15 hard code tasks at a fixed 20K-token budget, we compare four strategies: reasoning-only (extended thinking), best-of- N sampling with an oracle verifier, a single-agent loop (L), and the proposer-reviewer harness (H). Reasoning saturates at 73.3% (86.7% only under a thinking-heavy partition at $1.57\times$ the budget) and best-of- N at 86.7%, while *every* feedback-iterating strategy reaches $\geq 97.8\%$ (Figure 3).¹ Even against the *stronger* in-token-space axis (oracle best-of- N at 86.7%) the interaction gap is 11–13 pp; against reasoning-only at matched budget it is up to 27 pp. Either way the separation is the framework’s load-bearing prediction.

5.1 The active ingredient is execution, not review (P2)

Code as a clean case study. Code is the cleanest demonstration because the channel is binary and the score is objective. Single-shot pass-rate is 66.7% (10/15); the harness lifts it to a strict 100% in all three runs, recovering every first-shot failure with zero regressions. The mechanism is consistent across recovered traces: turn 1 emits code that compiles but mishandles an edge case (off-by-one boundaries, semantic-version comparison, CIDR arithmetic); turn 2 receives the `pytest` traceback—the failing assertion, the offending input, expected vs. actual—and rewrites the

¹Two numerical coincidences of this small suite, flagged so the tables read cleanly: the default-partition reasoning ceiling and the Phase-1 single-seed single-shot rate are both 73.3% (11/15); the thinking-heavy reasoning partition and the oracle best-of- N ceiling are both 86.7% (13/15). These are coincidences, not the same measurement; the canonical 3-seed single-shot rate is 66.7% (Table 17).

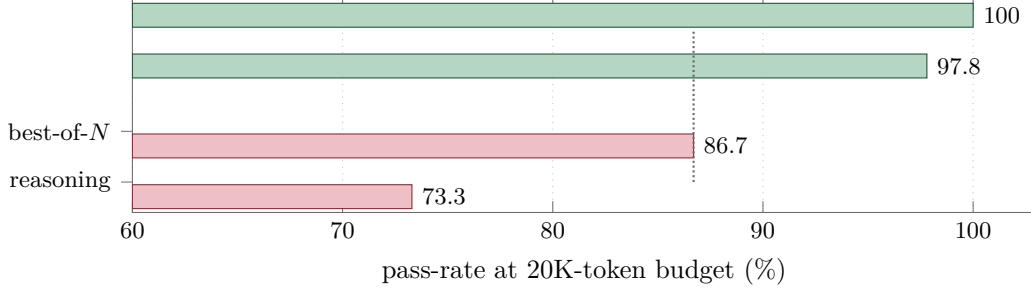


Figure 3: **Two axes saturate; the third does not.** At a matched 20K-token budget on hard code, the two in-token-space axes (red) plateau below the interaction strategies (green), which all reach $\geq 97.8\%$. The proposer-reviewer harness reaches strict 100% across seeds. Dotted line: best-of- N ceiling.

localized block; turn 3 confirms. Reasoning alone cannot supply this: the model has no way to know its boundary condition is wrong until a concrete test exercises it. A second, harder 11-task suite of deep-spec implementations (a from-scratch JSON parser, an A1 spreadsheet-reference resolver, a minimal edit-script diff, an arithmetic/boolean expression evaluator, ...; each validated against a reference solution) corroborates this at larger headroom: single-shot $69.7\% \rightarrow$ strict 100% , $+30.3$ pp (bootstrap 95% CI $[15.2, 45.5]$ pp), all 10 failing task-runs recovered, 0 regressions (sign-test $p = 0.002$).

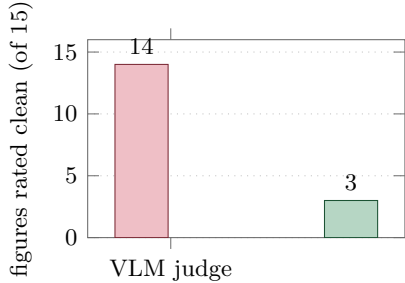
An in-modality control isolates execution. The cross-modal ordering could be confounded by task difficulty, so we fix the task suite (the 15 hard code tasks) and vary *only* the feedback channel: a Type 1 reviewer (LLM cross-review, no execution), a Type 2 reviewer (LLM + **ruff** static analysis, still no execution), and the Type 3a baseline. Forbidden from seeing test results, the Type 1/2 reviewers reach a reviewed ceiling of 13/15; the Type 3a (execution) reviewer reaches 14/15 and does so $\sim 2.5\times$ more token-efficiently (2,643 vs. $\sim 7,100$ tokens/task, terminating in 1.53 vs. 2.07 iterations). The separation is not the extra reviewing pass—both have one—but the *execution* of the tests.²

5.2 Architecture buys efficiency and reliability

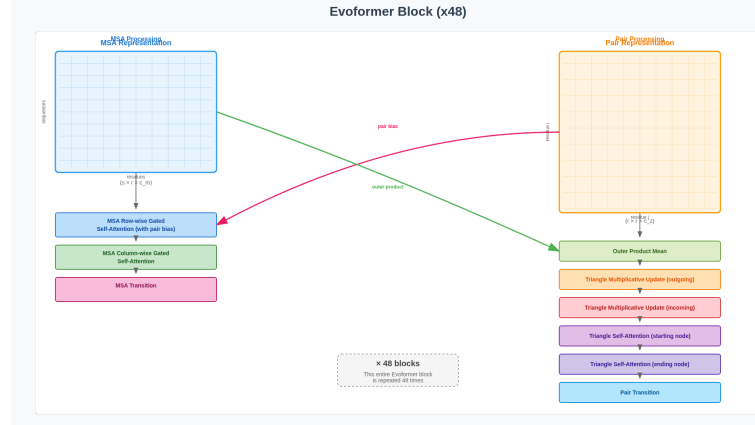
Efficiency. Among interaction strategies the gap is not in ceiling pass-rate (they tie within seed noise at $N=15$) but in cost and variance. The proposer-reviewer harness is the most token-efficient (1,029 mean output tokens vs. 1,431 for the single-agent loop and 1,416 for the oracle-verifier sample-and-select baseline) and the only variant at strict 100% with *zero* seed variance; the single-agent loop hits 100% in 2 of 3 seeds. Context isolation (the reviewer sees only what it needs to localize defects), structured critique (a typed defect list, not raw stderr), and role specialization explain the efficiency.

Built-in early stopping. The loop does not over-revise: on code, across three runs, *every* already-passing single-shot task is submitted at iteration 1 with no revision (30/30 instances, zero

²This Type 3a control is a single leaner-budget reference seed and caps at 14/15; its lone residual is `code_011` (CJK text wrapping), which the full harness at the 20K budget recovers in all three seeds—hence the strict 100% headline. Type 1 and Type 2 do not separate because the seeded bugs are runtime-logic errors **ruff** cannot see; a **mypy/semgrep**-catchable bug set is where the strict Type 2 > Type 1 prediction should be tested.



(a) The VLM passes 14/15; the deterministic instrument passes only 3/15.



(b) A single-shot figure the VLM rated “perfect”: both section titles are superimposed (“MSA Processing” over “MSA Representation”; “Pair Processing” over “Pair Representation”) and axis labels collide with the matrix borders.

Figure 4: **The default VLM judge is structurally blind to layout defects.** (a) On the same 15 single-shot academic figures, the VLM-on-a-screenshot judge rates 14 “perfect” while the deterministic DOM-geometry instrument finds only 3 actually clean; the other 11 are broken in ways cropped off-frame or below VLM acuity. (b) A representative “perfect”-rated render whose text-on-text overlaps (≥ 6 px) are exactly what the geometry check flags and the screenshot judge misses.

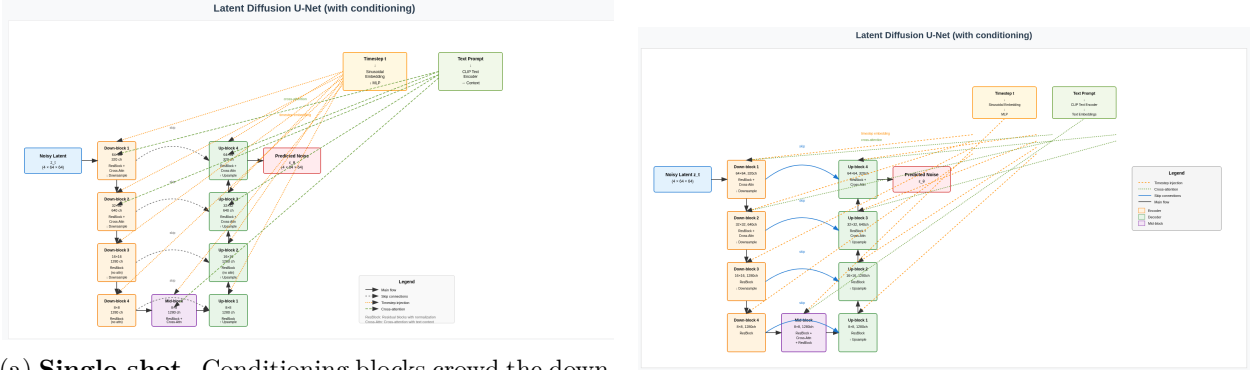
wasted tokens). The grounded signal—a failing test—is the trigger; without it the loop is silent. This is why code’s token overhead is only $\sim 37\%$ above single-shot despite an iteration cap of 5.

6 Grounding the Evaluation: The Gain Is Invisible to a VLM

The visual modalities expose the symmetric half of the framework (P3): even when grounded feedback *does* fix the artifact, an ungrounded metric cannot see it.

The default VLM judge is structurally blind. Our starting evaluator is a binary per-requirement rubric scored by a VLM over a screenshot. On 15 dense academic-paper architecture figures, even with native-resolution quadrant tiling, this judge rates 14 **of** 15 single-shot renders “perfect.” Yet by direct inspection they are broken—superimposed section titles, labels spilling out of boxes. The reason is mechanical: screenshots are captured at 1080 px, so any content overflowing the canvas is cropped *off-frame before the image reaches the judge*, and small in-frame overlaps fall below its acuity. A deterministic DOM-geometry check finds only 3 **of** 15 are actually clean (mean 1.6 defects). The VLM is blind to precisely the defects that matter (Figure 4a).

A deterministic instrument reveals a large, significant effect across four modalities. We score four visual modalities with the same DOM-geometry+alignment instrument under one identical configuration (Sonnet 4, temperature 0, design-principle prompt, three seeds, propose→measure→feed-exact-defects-back→revise, ≤ 3 iterations). Table 2 and Figure 6 report the result: grounded geometric feedback produces a large, significant reduction in real layout defects on *all four*, with bootstrap 95% CIs that exclude zero and improvements outnumbering regressions $\geq 5:1$. The magnitudes track single-shot headroom—dense responsive web pages are



(a) **Single-shot.** Conditioning blocks crowd the down-block column, edges route through boxes and tangle at the center, and the legend collides with the decoder—defects the geometry instrument flags but the VLM cannot see.

(b) **After grounded feedback.** The same task: conditioning blocks lifted clear, columns spaced and aligned, skip connections deconflicted, legend boxed in free space.

Figure 5: **What grounded geometric feedback actually fixes.** The same academic-figure task, single-shot (a) vs. after the propose→measure→revise loop (b). This is the per-artifact reality behind the -74% mean defect reduction in Table 2—and exactly the difference a screenshot judge is blind to.

Table 2: Deterministic geometric-feedback results, one identical configuration across four visual modalities (Sonnet 4, $T=0$, 3 seeds, alignment-inclusive reward). Δ is the mean defect reduction; CI is a paired bootstrap 95% interval on $\Delta\%$ (10k resamples); p is a two-sided paired sign test. Web is summed over 1920/375 widths; animations over sampled frames.

Modality	n	SS	Rev.	Δ	95% CI	impr./regr. (p)
Academic figures (20)	60	0.57	0.15	-74%	$[52, 89]\%$	17/2 (7×10^{-4})
Dense slides (12)	36	1.25	0.33	-73%	$[45, 93]\%$	13/1 (1.8×10^{-3})
Web pages (20)	60	16.1	8.5	-47%	$[30, 62]\%$	33/2 (4×10^{-8})
Animations (20)	60	16.9	10.2	-40%	$[10, 64]\%$	34/7 (3×10^{-5})

defect-rich out of the box (16 defects/page across desktop and mobile), while a frontier model under a design-principle prompt already lays out many figures and slides cleanly. Animations have the widest interval (a 40% reduction, CI $[10, 64]\%$): their per-task defect counts are heavy-tailed across the dynamic suite, so we read the sign test ($34/7$, $p = 3 \times 10^{-5}$) as the robust statement and the mean as indicative.

An ungrounded reviewer fails to fix layout—and on slides makes it worse (ablation, two modalities). Grounding matters on the feedback side too. We fix the task suite and the proposer model and swap *only* the reviewer’s grounding; because the two arms are separate runs, each is compared against its *own* single-shot baseline (which is why their single-shot values differ). The deterministic geometric-feedback reviewer sharply *reduces* mean defects (slides $1.25 \rightarrow 0.33$, figures $0.57 \rightarrow 0.15$), whereas the VLM-feedback reviewer does not reduce them and trends worse—slides $1.89 \rightarrow 2.44$ (geometrically-clean slides $16/36 \rightarrow 13/36$) and figures $0.52 \rightarrow 0.62$. The slide regression is the clear case; the figure shift is directional only (regressions outnumber improvements 10:5, two-sided sign test $p \approx 0.30$, not significant). The mechanism is that, blind to the true geometry, the VLM reviewer’s edits break alignment about as often as they fix it. On the same slides the

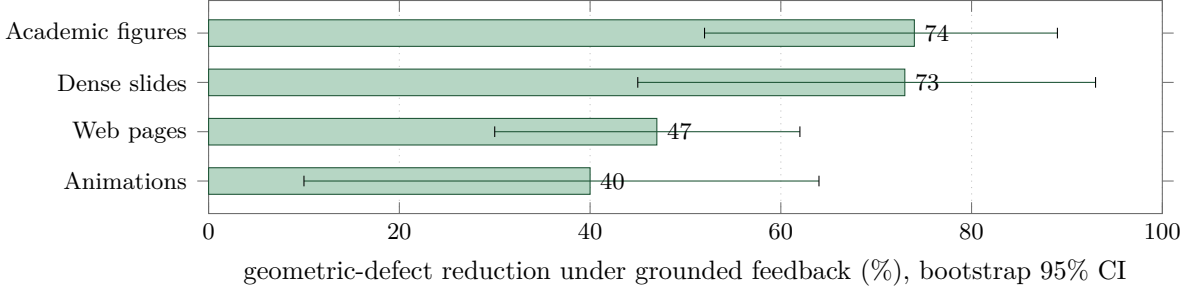


Figure 6: **Grounded geometric feedback removes real layout defects on all four visual modalities** (mean reduction; whiskers are paired-bootstrap 95% CIs, all excluding zero). The standard VLM-on-a-screenshot metric measures none of this.

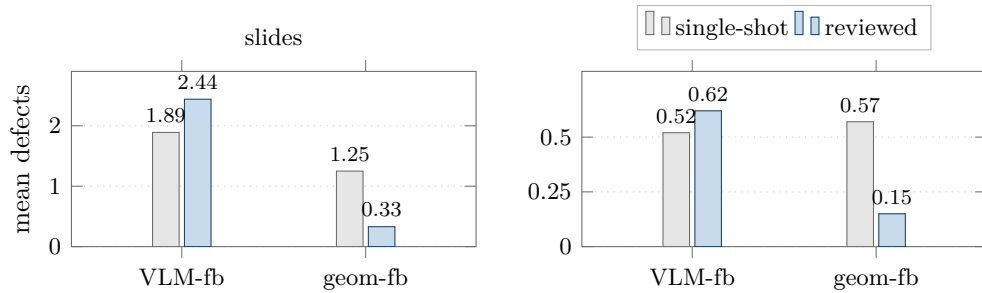


Figure 7: **An ungrounded reviewer fails to fix layout; a grounded one does (controlled ablation, two modalities)**. Same task suite and proposer model; only the reviewer’s grounding varies. The two arms are separate runs, so each is plotted against its *own* single-shot (hence the differing single-shot bars). The VLM-feedback reviewer (VLM-fb) does not reduce defects and on slides increases them (1.89 $\rightarrow$ 2.44; figures 0.52 $\rightarrow$ 0.62, directional/n.s.); the deterministic geometric-feedback reviewer (geom-fb) reduces them (1.25 $\rightarrow$ 0.33; 0.57 $\rightarrow$ 0.15). The grounded signal is necessary on the feedback side, not just the metric side.

binary VLM rubric simultaneously saturates (0.97 mean, 29/36 rated perfect), confirming the judge cannot even see the difference. Both halves of Figure 1 are necessary (Figure 7).

Is the reduction circular? The instrument both *drives* the revision and *scores* it, and the harness keeps the best-scoring iteration, so *some* reduction is mechanically guaranteed; the honest questions are how large it is and whether it reflects quality a human would care about. Three facts argue it is not merely an artifact of optimizing the reported number. **(i) The defects are real:** single-shot renders carry them at high rate (only 3/15 probe figures clean; Figure 4b), and the thresholds are deliberately conservative (≥ 6 px overlap, > 16 px overflow), so a flagged defect is visible, not sub-perceptual. **(ii) The magnitude is not preordained:** keeping the best of three iterations could in principle move the metric by almost nothing; that grounded revision instead removes 40–74% of defects is a property of the feedback, not the keep-best rule. **(iii) The ungrounded-reviewer ablation is the decisive control:** scored by the *same* metric, the VLM-feedback arm *fails* to reduce it (and worsens slides)—so the reduction tracks the *grounding* of the feedback, not the act of optimizing the scorer, a conclusion the model-free cross-model replication (Section 7) reinforces. What remains open—and is the principal caveat of the geometric measurement—is a quantitative human-preference study confirming that DOM-defect reduction maps onto perceived-quality gains

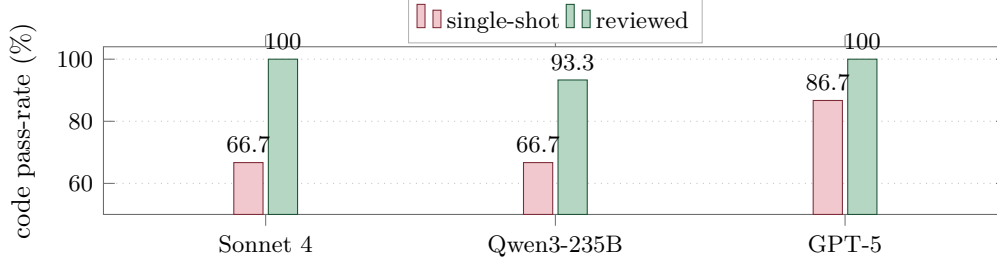


Figure 8: **The execution-grounded loop helps every model family.** Single-shot vs. harness-reviewed code pass-rate for three model families (+33.3/+26.7/+13.3 pp).

across the full suite.

Two modalities where the honest answer is “no headroom.” *Video editing*, scored by Gemini 3.1 Pro’s native full-clip rubric (a near-grounded instrument that observes pixels, not a still), is already strong single-shot (0.70, 22/45 task-runs perfect) so the reviewed lift is small and not significant (+0.04); a hardened multi-step suite de-saturates it (single-shot “fully correct” 49% → 11%). *Deep research*, scored against exact provided facts, saturates (0.982, +0.018, n.s.): a frontier model knows well-documented facts and does not take the planted traps, so the Type-3d lift is bounded by the near-zero single-shot error rate—a genuine limit of the factual modality, since de-saturating it would require facts the judge itself cannot grade. We report these as scoped negatives rather than headline lifts.

7 Robustness

Cross-model, code. The effect is architectural, not a Claude artifact. On the 15 hard code tasks the harness lifts Sonnet 4 by +33.3 pp (to 100%), Qwen3-235B by +26.7 pp, and GPT-5 by +13.3 pp (on a higher single-shot baseline; Figure 8). All three families benefit from the same execution-grounded loop.

Cross-model, grounded geometry. To rule out a Claude-specific or prompt-specific explanation of the grounded-evaluation effect, we run the same deterministic geometric-feedback harness with **Gemini 3.1 Pro** as the proposer on the dense real-paper slide suite, scored by the identical model-free DOM instrument. The effect reproduces and is in fact larger: mean defects fall 3.50 → 0.25 (−93%) over three seeds ($n=36$), geometrically-clean slides rise 17/36 → 34/36, and 19 of 20 decisive task-runs improve (sign-test $p = 4 \times 10^{-5}$). Gemini’s single-shot slides are *more* defect-prone than Sonnet’s (3.50 vs. 1.25), and the grounded loop removes almost all of it. Because the instrument depends on no model, this rules out a Claude-specific or scorer-specific explanation.

Held-out generalization. On a 32-task held-out code suite constructed after the method was fixed, the harness lifts 90.6% → 100%, recovering 3/3 first-shot failures with zero regressions—no tuning to the original tasks.

Budget allocation. Sweeping the proposer/reviewer token split on a fixed budget produces an 86.6 pp spread in pass-rate, monotone in the proposer’s per-call share: the optimum is *propose-heavy* (allocate the majority to generation, reserve enough for the reviewer to localize defects). Extra

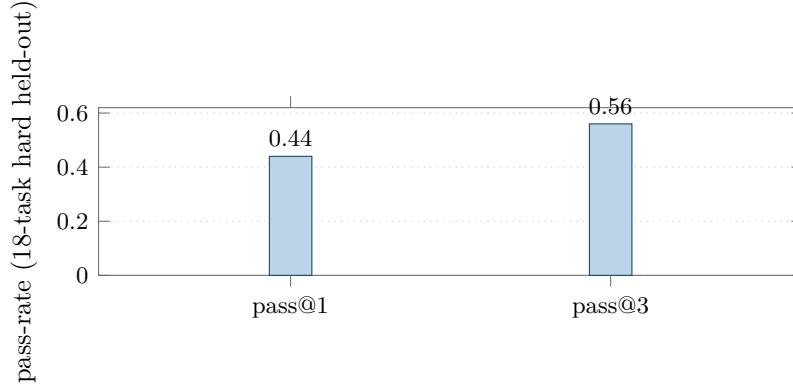


Figure 9: **An 8B student internalizes interaction quality.** Pass@1/pass@3 of the distilled 8B markdown student on the hard held-out suite ($0.51\times/0.70\times$ teacher mean@1/@2 on the 44-task OOD set; Table 12), at $\sim 10\times$ lower deployment cost.

inference compute is better spent on additional samples than on deeper iteration past the early-stop point.

8 Internalizing the Harness into a Small Student

Can the harness’s interaction-quality behavior be *internalized* so a small model emits high-quality artifacts in fewer passes? We distill judge-filtered teacher trajectories from the harness into an 8B Qwen3-VL student via SFT.

Result. The student reaches $0.51\times$ the teacher’s mean@1 capability on a 44-task out-of-distribution suite (Table 12; rising to $0.70\times$ at pass@2), and 44%/56% pass@1/pass@3 on an 18-task hard held-out suite, at $\sim 10\times$ lower deployment cost than running the harness over a frontier model (Figure 9). Run inside the same harness, the small student recovers more of the gap, confirming that scaffolding and internalization compose.

Variance is a budget (a cautionary finding). Reinforcement fine-tuning (RFT) on top of SFT improves every per-turn consistency metric we measure *and* regresses pass@3 by 17 pp. For sampling-based deployment, output-distribution variance *is* the inference-scaling lever, and any post-training that reduces it spends from the same budget best-of- N draws on. Variance is not noise to minimize; it is the resource sampling converts into quality. The operational recipe is therefore: SFT to internalize interaction quality, keep the proposer’s sampling temperature, and run it inside the grounded harness.

9 Discussion and Conclusion

Grounding is the load-bearing variable. Two of the three test-time axes are bounded by the data-processing inequality because they never leave the model’s token space; the third is unbounded precisely because grounded feedback imports external information each cycle. We have argued and shown that this grounding must hold on *both* sides of the loop. On the feedback side, an in-modality control isolates test *execution*—not the extra review pass—as the active ingredient, and a VLM *reviewer* of slide layout actively regresses quality. On the evaluation side, the default VLM-on-a-screenshot metric is structurally blind to the geometric defects the loop fixes, so the entire

visual-modality effect is invisible until the metric is grounded in deterministic DOM geometry—at which point grounded feedback removes 40–74% of real layout defects across four modalities, every CI excluding zero.

A methodological warning for the field. Much of the recent excitement about “LLMs that critique and improve their own visual output” is measured with VLM judges. Our results imply many such measurements are uninformative on layout: the judge cannot see the defects, and using it as the reviewer can make things worse. Where an artifact’s quality is a measurable physical property—geometry, execution, retrieval—the evaluation should be grounded in that property, not in a model’s picture of it.

Limits, honestly stated. Two modalities saturate: video editing is already strong single-shot (the lift is real only on a hardened multi-step suite), and deep research saturates because frontier models know well-documented facts and the judge cannot grade facts they do not. Deterministic geometry applies to artifacts with a measurable intended layout (figures, slides, web, animations); for genuinely flowing content we still rely on a binary rubric, with the VLM-reliability caveat. And because the geometric instrument both supplies the feedback and scores the result, the reported defect reductions are partly an optimization of the metric itself; we argue in Section 6 (via conservative thresholds, the non-trivial magnitude, and the ungrounded-reviewer control that moves the same metric the wrong way) that they nonetheless reflect real, human-visible defects, but a quantitative human-preference validation remains future work.

Future work and broader impact. Two directions follow directly. First, deterministic instruments for modalities beyond layout and execution—differentiable or symbolic checks for audio, 3D, tabular, and UI-interaction artifacts—would extend grounded evaluation to where VLM/LLM judges are likewise blind. Second, our DPI argument bounds an information channel; tightening it into a quantitative relationship between feedback fidelity and achievable quality is open. On impact: grounding the metric is a cheap, deployable correction to a measurement error that, we argue, currently inflates reported progress on self-improving visual generation; the same instruments are auditable and reduce reliance on opaque model judges. The harness itself is scaffolding over a frozen model, so it inherits that model’s capabilities and failure modes rather than introducing new ones.

Operational recipe. Wrap a frozen frontier model in a proposer–reviewer harness with the most actionable grounded feedback channel for the modality. For visual artifacts, ground *both* the reviewer and the scorer in deterministic geometry—measure the rendered DOM, never a screenshot. Allocate the majority of the per-task budget to the proposer; spend extra compute on samples, not deeper iteration. For cost-sensitive deployment, distill the proposer into a small student and run it inside the same harness. Interaction scaling is real, distinct from reasoning and sampling, predictable from the feedback taxonomy, and—once you ground the metric—plainly visible.

References

- [1] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling, 2024.

- [2] Stephen Casper, Xander Davies, Claudia Shi, Thomas Krendl Gilbert, et al. Open problems and fundamental limitations of reinforcement learning from human feedback, 2023.
- [3] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.
- [4] DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning, 2025.
- [5] Jonas Gehring, Kunhao Zheng, Jade Copet, Vegard Mella, Taco Cohen, and Gabriel Synnaeve. RLEF: Grounding code LLMs in execution feedback with reinforcement learning. In *International Conference on Machine Learning*, 2025.
- [6] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *International Conference on Learning Representations*, 2024.
- [7] Dingkun Hong. The practice and exploration of AI coding. Keynote, ByteDance/Volcano Engine Force Conference, Beijing; speaker is VP of Engineering at ByteDance, 2026. <https://mp.weixin.qq.com/s/tJAinVKzZAqZrhiEvGU2Dg>.
- [8] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations*, 2024.
- [9] Ryo Kamoi, Yusen Zhang, Nan Zhang, Jiawei Han, and Rui Zhang. When can LLMs actually correct their own mistakes? a critical survey of self-correction of LLMs. *Transactions of the Association for Computational Linguistics*, 2025.
- [10] Zimu Lu, Houxing Ren, Yunqiao Yang, Ke Wang, Zhuofan Zong, Junting Pan, Mingjie Zhan, and Hongsheng Li. WebGen-Agent: Enhancing interactive website generation with multi-level feedback and step-level reinforcement learning, 2025.
- [11] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Sean Welleck, Bodhisattwa Prasad Majumder, Shashank Gupta, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, 2023.
- [12] Shishir G. Patil, Tianjun Zhang, Xin Wang, and Joseph E. Gonzalez. Gorilla: Large language model connected with massive APIs. In *Advances in Neural Information Processing Systems*, 2024.
- [13] Yangjun Ruan, Eleftheria Briakou, Charles Xie Han, Yu Chen, Yufan Jiao, et al. Iterative agent decoding for detecting compounding errors in LLM agents, 2025.
- [14] Timo Schick, Jane Dwivedi-Yu, Roberto Dessí, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, 2023.
- [15] Junhong Shen, Hadi Jain, Zelai Xiao, Brandon Amos, Aaditya Ramdas, Yuandong Tian, and Alborz Geramifard. Thinking vs. doing: Agents that reason by scaling test-time interaction. In *Advances in Neural Information Processing Systems*, 2025.

- [16] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, 2023.
- [17] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. In *International Conference on Learning Representations*, 2025.
- [18] Kimi Team. Kimi k1.5: Scaling reinforcement learning with LLMs, 2025.
- [19] Dat Tran and Douwe Kiela. Single-agent LLMs outperform multi-agent systems on multi-hop reasoning under equal thinking token budgets, 2026.
- [20] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. In *NeurIPS Foundation Models for Decision Making Workshop*, 2023.
- [21] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.
- [22] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022.
- [23] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, 2023.
- [24] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.
- [25] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

A Detailed Results and Configurations

This appendix collects the full tables underlying the summaries in the main text: the four-strategy scaling curves, the matched-budget reasoning baseline, the in-modality feedback-type control, the single-agent-vs-harness comparison, cross-model and held-out code, the budget-allocation simplex, token ROI by modality, the distillation ablations, the out-of-distribution capability ratio, the per-task pass@ k breakdown for the 8B student, and the original six-modality holistic overview (superseded for the visual modalities by the grounded measurements of Section 6).

Table 3: **The headline scaling result (3-seed mean $\pm$ SD).** Pass-rate by strategy $\times$ budget on 15 hard code tasks, Sonnet 4. S, L, and H are averaged over three independent on-policy runs; R was run once (extended-thinking variation across seeds is small at temp 0) at the sweep’s budget partition (a more generous partition lifts R to 86.7%; see footnote and Table 4). S’s small dip from $B=1K$ (80.0%) to $B=5K$ (77.8%) is within the ± 3.1 pp seed noise on a 3-seed $\times$ 15-task suite; pass@ N is non-decreasing in N in expectation but not pointwise across finite-sample on-policy seeds. *Reasoning-only and best-of- N saturate well below 100%*; all three iteration-with-feedback strategies reach $\geq 97.8\%$ at $B=20K$ (L hit 100% on 2 of 3 seeds; H on all 3; H is the only variant at strict 100% across seeds). The architectural ordering across H, L is within seed noise on pass-rate; what separates them is token efficiency and reliability (Section 5).

Budget B	R (reasoning-only)	S (best-of- N)	L (single-agent loop)	H (proposer-reviewer)
1K	60.0%	80.0% ± 0.0 pp	57.8% ± 3.1 pp	62.2% ± 3.1 pp
5K	73.3%	77.8% ± 3.1 pp	93.3% ± 0.0 pp	91.1% ± 3.1 pp
20K	73.3%	86.7% ± 0.0 pp	97.8% ± 3.1 pp	100.0% ± 0.0 pp
Ceiling	73.3%	86.7%	$\sim 100\%$	100% (zero variance)

Table 4: Reasoning-only at matched budget vs. single-shot and the harness on the 15 hard code tasks. Single-seed numbers (the Phase-1 reference seed, also used in Table 5); the 3-run aggregate is reported in Table 17 ($66.7 \pm 6.7\%$ single-shot, $100.0 \pm 0.0\%$ reviewed). *Reading the numbers:* single-shot here is this single seed’s $11/15 = 73.3\%$ (the canonical 3-seed mean is 66.7%), and the reasoning value shown is the *thinking-heavy* partition (86.7%); this coincides numerically with the oracle best-of- N ceiling, while the *default*-partition reasoning ceiling (73.3%, Table 3) coincides with this seed’s single-shot—both are coincidences of the small suite, not the same run. Reasoning closes two-thirds of the harness’s gap over single-shot but falls 6.7 pp short of the harness’s pass-rate even at $1.57\times$ the harness’s token budget.

Strategy	Pass-rate	Mean tokens	Δ vs. single-shot
Single-shot	73.3% (11/15)	1,406	—
Reasoning-only (thinking-heavy partition)	86.7% (13/15)	4,137	+13.3 pp
Proposer-reviewer harness	93.3% (14/15)	2,643	+20.0 pp

Table 5: In-modality feedback-type controls on 15 hard code tasks (Sonnet 4 proposer and reviewer). Each row is a single on-policy seed; the “SS” column is the *per-seed* single-shot pass-rate (not held constant across rows) and “Reviewed” is the ceiling after that seed’s harness run. The three SS values (10/15 and 11/15) are draws from the same single-shot distribution that gives Table 17’s 3-run mean of $66.7 \pm 6.7\%$ (i.e. within the ± 1 SD band centered on 10/15). Because the SS column varies by row, the load-bearing comparison is the *reviewed* ceiling: Type 3a reaches 14/15, Type 1/2 reach 13/15 (+6.7 pp), and Type 3a is $\sim 2.5\times$ more token-efficient (2,643 vs. $\sim 7,100$ tokens/task; 1.53 vs. 2.07 iterations). The single task unrecovered in every control row is `code_011` (CJK text wrapping); the full harness at $B=20K$ recovers it across all three seeds (Table 3), which is why the headline reviewed rate is the strict 100% of Table 17 while this leaner single-seed control caps at 14/15.

Type	Reviewer sees	SS	Reviewed	Iters
None (single-shot)	—	11/15	—	—
Type 1 (cross-review)	code only	10/15	13/15	2.07
Type 2 (+ruff)	code + static analysis	11/15	13/15	2.07
Type 3a (+exec)	code + test stderr/traceback	11/15	14/15	1.53

Table 6: Three interaction-strategy variants at $B=20K$ on the 15 hard code tasks, all using Type 3a execution feedback. Pass-rate is within seed noise (sign-test $p>0.6$ on the H–L comparison); the harness wins on *token efficiency* (1,029 vs. 1,431 L vs. 1,416 IAD mean output tokens) and on *seed variance* (zero across three seeds for H; L hit 100% in 2 of 3 seeds; IAD was run for one seed only).

Strategy	Pass-rate	Mean tokens	Notes
L (single-agent loop)	97.8% ± 3.1 pp (3 seeds)	1,431	1 task in 1 seed
IAD (oracle, K=3, T=0.7)	100.0% (1 seed)	1,416	+38% tokens vs. H
H (proposer–reviewer)	100.0% ± 0.0 pp (3 seeds)	1,029	zero seed variance

- A.1 Four-strategy scaling at a matched 20K-token budget
- A.2 Reasoning-only at a matched budget
- A.3 In-modality feedback-type control (Type 1/2/3a) on a fixed code suite
- A.4 Single-agent loop vs. proposer–reviewer harness
- A.5 Cross-model replication (code)
- A.6 Held-out generalization (code)
- A.7 Budget-allocation simplex
- A.8 Token ROI by modality
- A.9 Distillation: headline
- A.10 Distillation: out-of-distribution capability ratio
- A.11 Distillation: capability U-curve over force-close budget
- A.12 Distillation: data-scale
- A.13 Distillation: RFT regresses pass@3
- A.14 Per-task pass@ k for the 8B markdown student
- A.15 Original six-modality holistic overview (historical)

Table 7: Cross-model replication on the same 15 hard code tasks. The Claude row is the multi-run reference from Table 17; Qwen and GPT-5 are single on-policy runs at $T=0.7$. The harness lift survives across three families (Anthropic, Alibaba, OpenAI).

Model	Single-shot pass	Reviewed pass	Δ (lift)
Claude Sonnet 4 (3-run mean)	$66.7 \pm 6.7\%$	100.0 $\pm$ 0.0%	+33.3 pp
Qwen3-235B-Instruct-2507 (1 run)	66.7% (10/15)	93.3% (14/15)	+26.7 pp
GPT-5 (1 run)	86.7% (13/15)	100.0% (15/15)	+13.3 pp

Table 8: Held-out generalization on 32 newly constructed code tasks (zero overlap with the 15-task development set). The harness recovers 3/3 single-shot failures and regresses zero passing tasks. The smaller absolute Δ is a baseline-compression effect (single-shot is 90.6% on the held-out set, leaving at most 9.4 pp headroom).

Split	N	Single-shot	Reviewed	Δ	SS-fail fixes / regressions
Development (3-run mean)	15	$66.7 \pm 6.7\%$	$100.0 \pm 0.0\%$	+33.3 pp	5/5 / 0
Held-out v2	32	90.6%	100.0%	+9.4 pp	3/3 / 0

Table 9: Budget allocation simplex on 15 hard code tasks, $B = 10K$ total. Pass-rate is monotone in the proposer’s per-call cap; review-heavy corners (low b_1) collapse. The 9-point sweep yields an 86.6 pp spread.

Allocation	b_1 (prop)	b_2 (exec)	b_3 (rev)	Pass-rate	Mean tokens
A (propose-heavy)	0.80	0.10	0.10	93.3% (14/15)	3,101
G (prop-dominant)	0.50	0.25	0.25	93.3% (14/15)	2,697
D (prop+exec)	0.40	0.40	0.20	80.0% (12/15)	4,025
E (prop+review)	0.40	0.20	0.40	80.0% (12/15)	4,388
I (equal)	0.33	0.34	0.33	66.7% (10/15)	4,845
H (review-dominant)	0.25	0.25	0.50	40.0% (6/15)	7,485
F (exec+review)	0.20	0.40	0.40	13.3% (2/15)	8,771
B (execute-heavy)	0.10	0.80	0.10	6.7% (1/15)	9,383
C (review-heavy)	0.10	0.10	0.80	6.7% (1/15)	9,383
Spread:				86.6 pp	

Table 10: Per-task token cost and quality return-on-investment for the harness, averaged across three on-policy runs. “Extra tokens” is reviewed-minus-single-shot per task. Tokens per 0.01 quality is extra tokens divided by (mean reviewed quality – mean single-shot quality)·100. Code dominates by 100 $\times$.

Modality	Single-shot tokens	Reviewed tokens	Extra tokens	Tokens / 0.01 quality
Code	3,336	4,575	1,239	37
Video	3,808	20,584	16,776	359
Research	14,116	27,793	13,677	2,415
Webpages	14,060	80,148	66,088	5,128
Animations	25,491	89,009	63,518	4,331
Slides	10,274	36,956	26,682	4,447

Table 11: Distilled 8B student on the 18-task hard held-out suite (teacher’s pre-curated single-shot failures, so teacher single-shot is 0/18 by construction). The student lifts judge-keep from 0% to 44% at pass@1 and 56% at pass@3. The $0.51\times$ mean@1 capability ratio against the teacher is established on a separate 44-task out-of-distribution suite (Table 12), where teacher single-shot is non-degenerate.

Setup	Compute	Judge-keep
1 sample $\times$ 5 turns	1 $\times$	44% (8/18)
1 sample $\times$ 8 turns	1.6 $\times$	28% (regressed)
3 samples $\times$ 5 turns (pass@3)	3 $\times$	56% (10/18)

Table 12: Distilled 8B student (V3 + force-close) on a 44-task out-of-distribution suite where the teacher’s single-shot is non-degenerate (teacher 20/44). Capability ratio is the student’s judge-keep divided by the teacher’s 20/44; two sampling seeds, pass@2 is the union over both. This is the suite that establishes the $\sim 0.51\times$ mean@1 / $0.70\times$ pass@2 ratios quoted in Section 8 and Figure 9.

Metric	sample 1	sample 2	mean@1	pass@2
Judge-keep	9/44 (20%)	11/44 (25%)	23%	32%
Capability ratio (vs teacher 20/44)	0.45 $\times$	0.55 $\times$	0.51 $\times$	0.70$\times$

Table 13: Reasoning-cap U-curve on the 18-task hard held-out suite. SFT recipe sweep on Qwen3-VL-8B-Instruct (37 judge-kept teacher traces; same inference protocol). The optimum is at the student’s coherence horizon, not the teacher’s.

Variant	Reasoning cap (chars)	Judge-keep	Identical-retry
V1 (no reasoning)	0	6%	67%
V4	1000	11%	35%
V2	3000	17%	32%
V3 (chosen)	1500	44%	10%

Table 14: Three independent attempts to scale the SFT corpus past V3’s 37 examples. All regressed. Quality dominates quantity at this scale.

Variant	Composition	n examples	Judge-keep
V3 (chosen)	235B teacher, judge-kept	37	44%
V5	V3 + 13 judge-rejected (235B)	50	33%
V6	V3 + 20 judge-kept (30B teacher)	57	33%
RFT v1	Self-rolled, judge-filtered	61	39%

Table 15: RFT polishes consistency at the cost of pass@ k . Per-turn metrics improve; per-task variance collapses; pass@3 regresses by 17 pp. For deployment with sampling, the model’s variance *is* the inference-scaling budget. *Note on notation.* **mean@1** is the expected pass-rate of a single random sample, computed as the mean accuracy across the three sampling seeds (one $T=0$ greedy + two $T=0.7$). This is the natural variance-sensitive metric and differs from the greedy single-seed $pass@1=44\%$ reported in Table 11, which is just seed 1’s accuracy. $pass@2$ and $pass@3$ are union pass-rates over the first 2 / all 3 seeds.

	Consistency metrics			Sampling-aware judge-keep		
	id-retry ↓	addresses-rev ↑	no-artifact ↓	mean@1	pass@2	pass@3
V3 (chosen)	9%	83%	2	31%	50%	56%
RFT v1	8%	89%	1	28%	39%	39%

Table 16: Per-task judge-keep outcome for the 8B markdown student on the 18-task hard held-out suite. ✓ = judge-kept; · = judge-rejected.

Category	Task	seed 1 ($T=0$)	seed 2 ($T=0.7$)	seed 3 ($T=0.7$)	pass@3
webpages	web_002	✓	·	·	✓
	web_003	✓	·	✓	✓
	web_005	·	·	·	·
	web_008	·	·	·	·
	web_012	·	·	·	·
slides	slide_001	·	·	·	·
	slide_006	·	·	·	·
	slide_007	·	·	·	·
	slide_008	✓	·	·	✓
	slide_010	·	·	·	·
	slide_014	✓	✓	✓	✓
	slide_018	✓	✓	·	✓
	slide_020	·	✓	·	✓
code	code_h001	✓	·	·	✓
	code_h002	✓	✓	·	✓
	code_h003	·	·	·	·
	code_h004	·	·	✓	✓
	code_h005	✓	✓	✓	✓
Total kept:		8/18	5/18	4/18	10/18 (56%)

Table 17: Single-shot vs. reviewed quality across six modalities, Claude Sonnet 4. Means and standard deviations are computed across three independent on-policy runs (per-run means, then aggregated). Code is binary pass-rate; other modalities are continuous in $[0, 1]$. p -values: one-sided paired sign test on per-task means. **Bold** marks rows where the lift clears $p < 0.05$. Code is scored by execution (`pytest`); video by a native full-video rubric (Gemini 3.1 Pro judging the whole rendered clip); the four visual/factual rows use a holistic screenshot judge and serve only as a cross-modal overview. *This table is historical and is superseded for the visual modalities by the grounded measurements of Section 6:* under the deterministic DOM-geometry instrument (Table 2) the figure, slide, web, and animation effects are large and significant, whereas the holistic-judge numbers here cannot see the geometric defects and therefore mis-state them. The visual rows here also use the earlier, smaller task sets (15 web and 15 animation tasks, 20 generic slides), not the hardened suites of Table 2 (20/20/12 real-paper); this is why the per-modality N differs between the two tables. The deep-research and video rows also use *different* instruments than Section 6: research here is the old holistic judge ($0.63 \rightarrow 0.69$) rather than the search-grounded factual rubric (single-shot 0.982), so the large numeric gap is an instrument change, not a contradiction; video uses the same Gemini-native rubric in both places ($0.70 \rightarrow 0.74$).

Modality	N	Feedback type	Single-shot	Reviewed	Δ (sign-test p)
Code	15	3a	$66.7 \pm 6.7\%$	$100.0 \pm 0.0\%$	+33.3 pp (0.008)
Video editing	15	3a+3c	0.70 ± 0.03	0.74 ± 0.05	+0.04 (0.64)
Deep research	15	3d	0.63 ± 0.06	0.69 ± 0.06	+0.06 (0.090)
Web pages	15	3b	0.43 ± 0.03	0.56 ± 0.01	+0.13 (0.073)
Animations	15	3c	0.40 ± 0.03	0.55 ± 0.01	+0.15 (0.073)
Slide decks	20	3b	0.73 ± 0.01	0.79 ± 0.01	+0.06 (0.090)